

# Neural Networks G5015

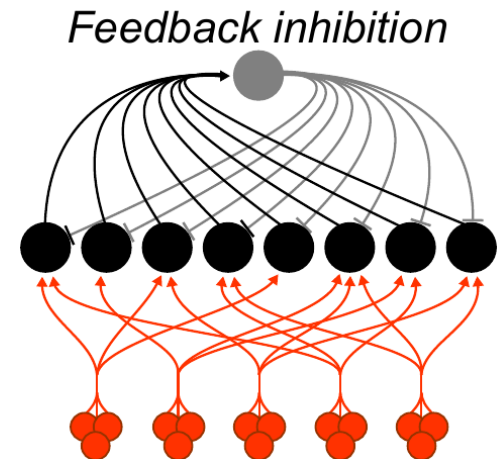
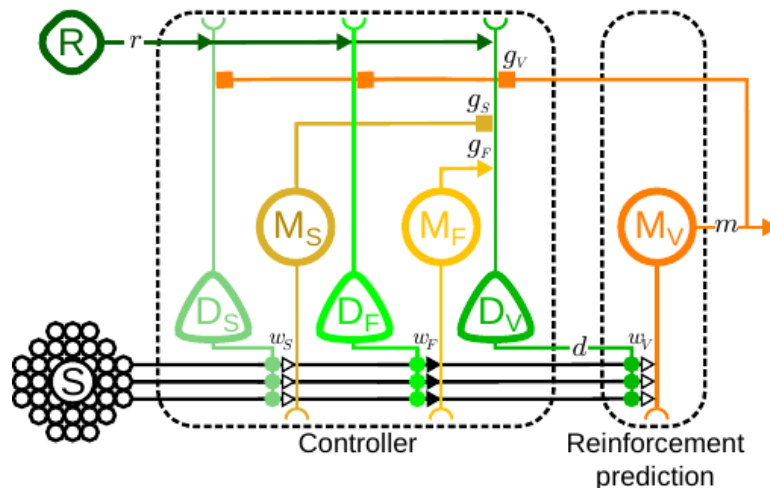
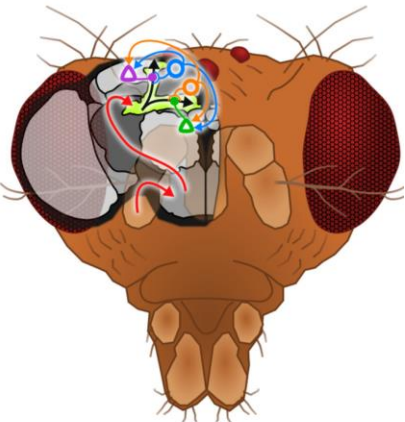
## Lecture 1



[pollev.com/jbennettlecture](https://pollev.com/jbennettlecture)

# Dr. James Bennett

- My group is interested in neural algorithms
  - in the brain (especially in insects), e.g.
    - how might biological neural networks implement algorithms often used in machine learning, e.g. the delta rule in reinforcement learning?
    - what are they doing that's different to ML algorithms?
  - in artificial neural networks, e.g.
    - how can properties of biological neural circuits be incorporated into artificial neural networks to improve their performance?
    - multi-objective optimisation, different types of neurons carry special functions.



# Module Info

## Teaching:

- **Lectures:** James Bennett, [james.bennett@sussex.ac.uk](mailto:james.bennett@sussex.ac.uk), Lecture 11 given by Prof. Luc Berthouze.
- **Labs:** Yue Zhao, Isabel Lahmann

## Timetable

- **Lecture:** Weds 09:00-11:00 in Pevensey 1 1A7
- **Labs:**
  - Group 1 – Thurs 11.00-12:00, Chi. 1 rooms 16-18
  - Group 2 – Thurs 18:00-19:00, Chi. 1 rooms 14-15

## Q&A (Office hours):

Wed 11:00-13:00. Always best to schedule in advance using Outlook Calendar Invite.

**Discussion forum (Canvas)** General Q&A, Assessment Q&A

# Module content

- Fully Connected Neural Networks

- Machine Learning basics (Linear models)
- Perceptron, Multi-layer Perceptron (MLP)
- Backpropagation
- Gradient Descent optimisation

Weeks 1-4

- Deep Convolutional Neural Networks (CNNs)

- CNN basics (filters, pooling, efficiency)
- Advanced CNN architectures
- Model validation, gradient flow, regularisation
- Generative models (GANs)
- Sequential Models (RNNs, LSTMs)

Weeks 5-9

- Advanced Topics

- Graph Neural Networks
- Reinforcement learning

Weeks 10-11

# Lab sessions

## We'll be using:

- Python
- Pytorch
- Jupyter Notebooks
- Google Colab

## Labs will involve:

- Coding exercises
- Experimentation with code
- Thought exercises
- Potentially a journal club session (work in progress!)

DTs are there to help guide you through the material and discuss topics/challenges you're facing. Make the most of their help!

# Assessment

- 100% coursework based.
- You will be expected to perform and write up Methods and Results for a set of experiments. Your coursework should:
  - Demonstrate your ability to read, summarise, and critique academic literature relevant to NN techniques (peer-reviewed articles, books etc.)
  - Demonstrate your ability to design, optimise, test, and analyse your own NN and apply it to specified tasks.
  - Communicate ideas about NN techniques clearly, present data you obtain from your NN in a clear and concise way.
  - Describe advantages/disadvantages of the techniques you have used, as well as of techniques you didn't use, and to provide reasons (citing literature where relevant) why you made particular design choices, and why other approaches might provide an improvement.
  - Word limit of approx. 2000-3000 words.
  - **Report due: 12th May, 16.00.**
- **I will try to highlight important considerations for your assessment throughout the lectures!**

# Learning outcomes

By the end of this module, we expect students to be able to:

- demonstrate knowledge of the fundamental principles and limitations of linear and non-linear models (Regression, Perceptron, Support Vector Machines).
- demonstrate a detailed understanding of Multi-Layer Perceptrons (MLP), and modern deep learning frameworks such as Convolutional neural networks (CNNs).
- explain the fundamental techniques of backpropagation, including the math behind neural network training.
- develop and apply neural networks to real-world problems.
- communicate reasons for and against the use of a given technique when developing a neural network to solve a problem.

# Suggested reading

- ❑ [Deep Learning](#) (2016), Ian Goodfellow, Yoshua Bengio, Aaron Courville
- ❑ [Dive into Deep Learning](#) (2023), Aston Zhang, Zachary Lipton, Mu Li, Alexander Smola
- ❑ [Pattern Recognition and Machine Learning](#) (2006), Christopher Bishop
- ❑ [Understanding Deep Learning](#) (2023), Simon Prince



# Only one rule: Ask Questions!

PLEASE ask questions if anything is ambiguous, incorrect, or just interesting to you!.....

.....Please!

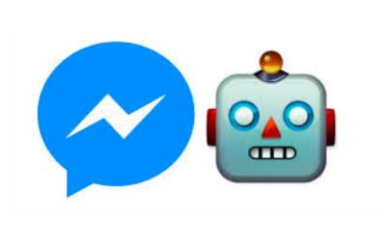
Poll Everywhere: ask questions anonymously during lectures using the pinned activity, "**Lecture Questions**".

Use the Discussion forum on Canvas to ask questions: chances are likely that someone else in the class would also like to hear the answer!

# Content: Today

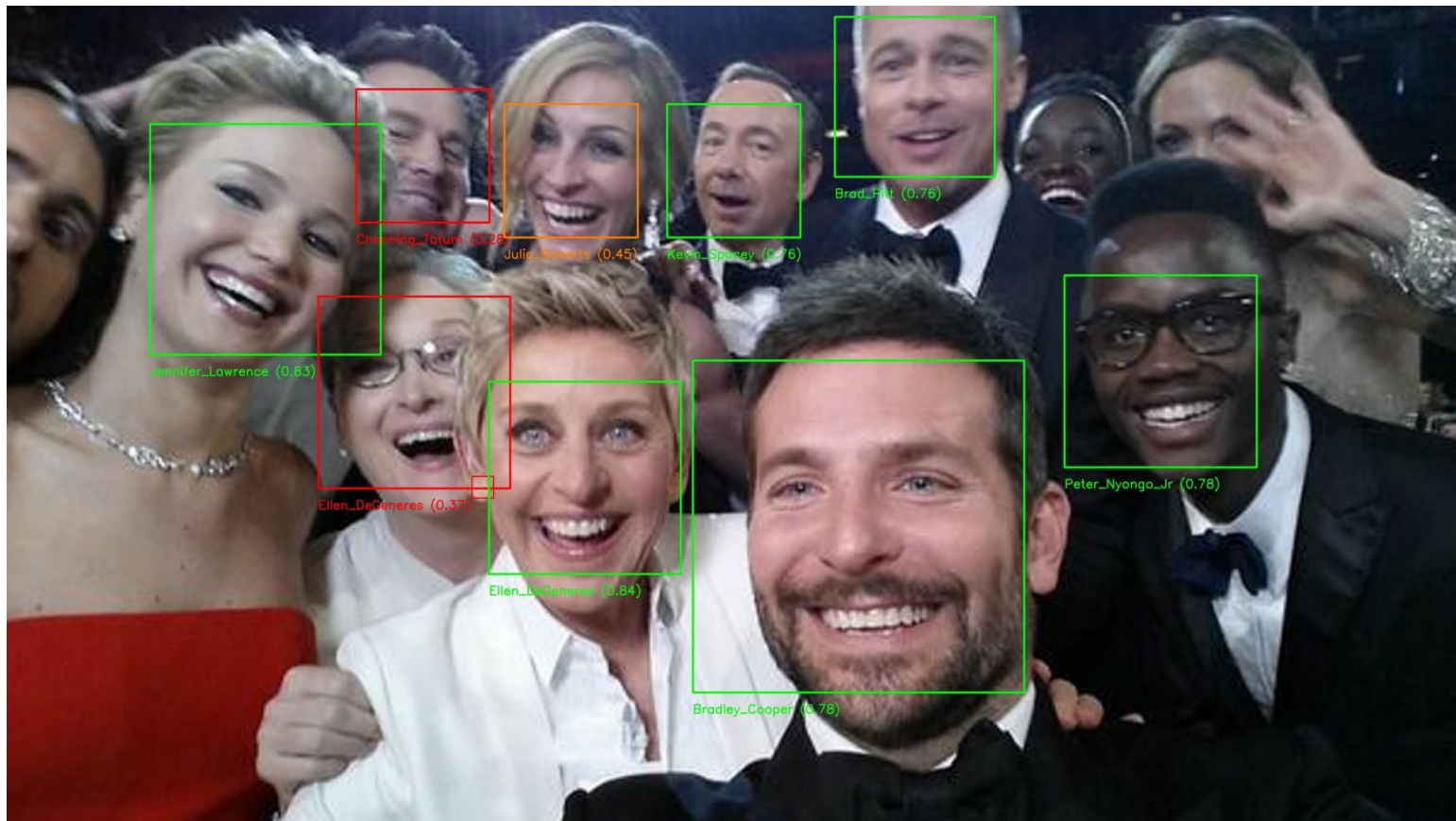
- **Intro**
- Machine Learning basics
  - Linear regression
  - Classification
- Extra: Linear Algebra and Calculus

# Deep learning is widely used today



# Example deep learning tasks

- Face recognition



- Machine translation





- Intelligent games





# Medical imaging



**SCIENTIFIC** REPORTS

Article | [Open Access](#) | [Published: 29 August 2019](#)

## **Deep Learning to Improve Breast Cancer Detection on Screening Mammography**

[Li Shen](#) , [Laurie R. Margolies](#), [Joseph H. Rothstein](#), [Eugene Fluder](#), [Russell McBride](#) & [Weiva Sieh](#)



# Example deep learning tasks

- Fake news generation



# Example deep learning tasks

- Speech recognition



**Bloomberg  
Businessweek**

May 21, 2018

THE SOONER  
THAN YOU  
THINK ISSUE



## *A.I. Painted This*

*It's also driving cars, making money, exploring  
oceans...and freaking people out*





# Style transfer



Artistic style transfer

# Machine Learning



WIKIPEDIA  
The Free Encyclopedia

## Machine learning

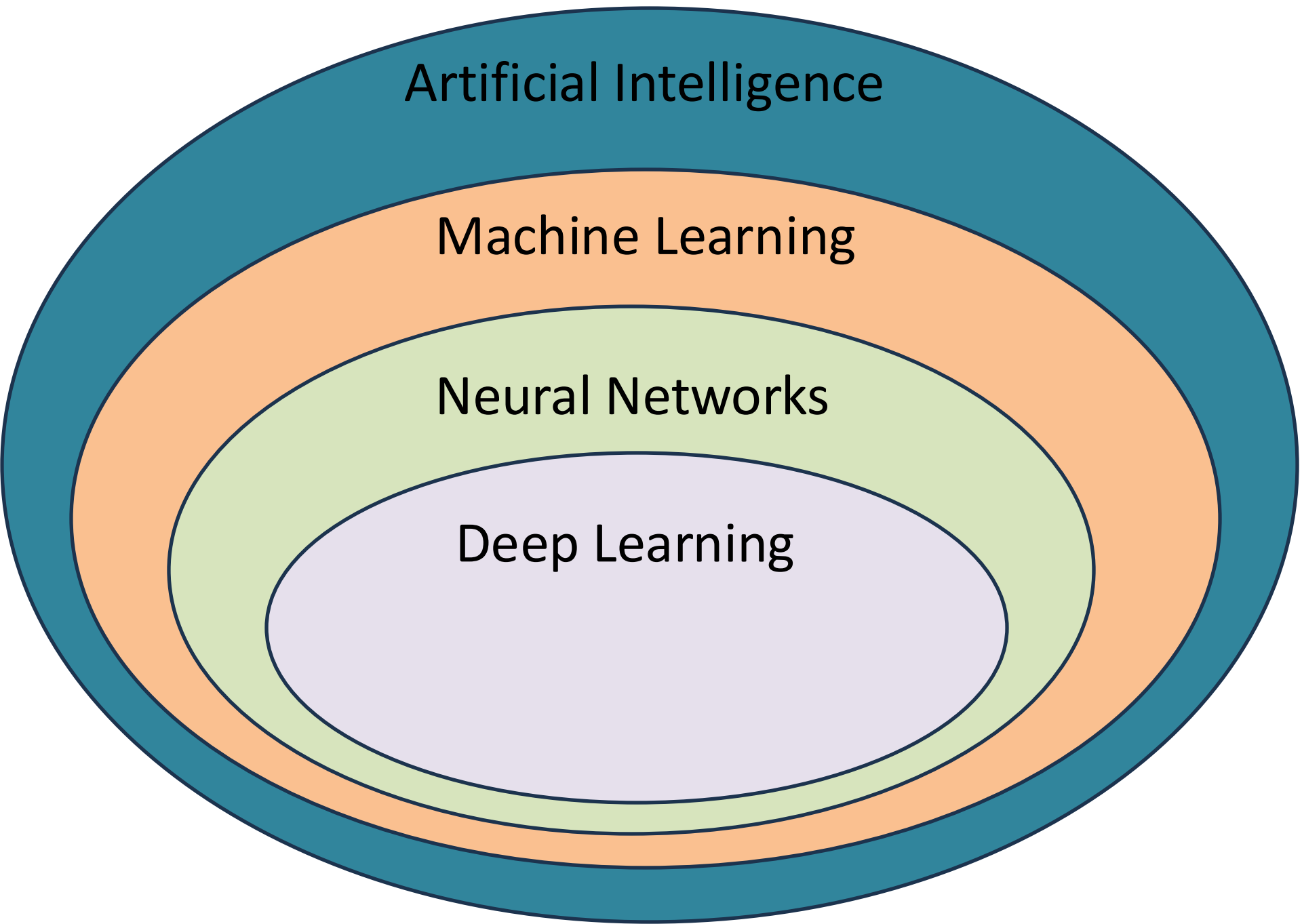
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[Article](#) [Talk](#)

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From Wikipedia, the free encyclopedia

**Machine learning (ML)** is a [field of study](#) in [artificial intelligence](#) concerned with the development and study of [statistical algorithms](#) that can learn from [data](#) and [generalize](#) to unseen data, and thus perform [tasks](#) without explicit [instructions](#).<sup>[1]</sup> Within a subdiscipline in machine learning, advances in the field of [deep learning](#) have allowed [neural networks](#), a class of statistical algorithms, to surpass many previous machine learning approaches in performance.<sup>[2]</sup>



# What is deep learning?

- One approach to finding patterns and relationships in data.
- Finding the right representations of the data to enable good performance on a task.
- The approximation of a very complex computation, achieved by performing relatively simple computations many times in parallel and stacking those computations.
- Examples of tasks that we get neural networks to learn:
  - Predict a category (label) for an image
  - Translate text from one language to another
  - Predict the trajectory of a moving target



# What is deep learning?

- One approach to finding patterns in data
- Finding the right representation of the data to enable good performance on a task.

Recent buzzword  
(concepts are ~60 years old!)

- The approximation of a very complex task by performing relatively simple computations in parallel and stacking those computations.

A class of ML methods that work remarkably well in some AI tasks

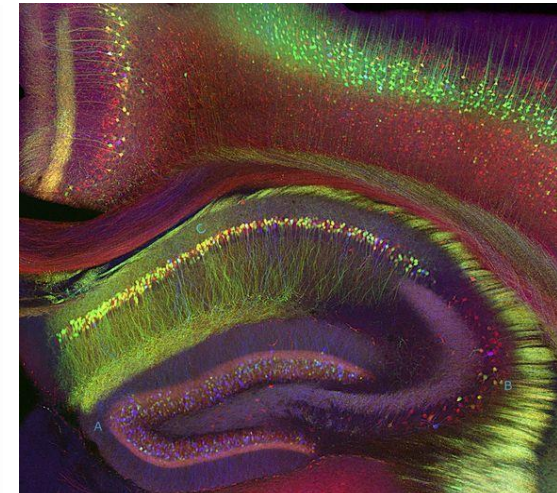
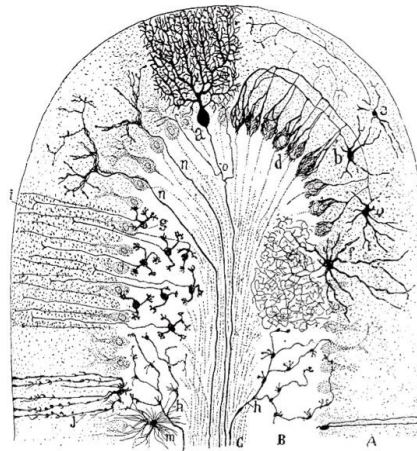
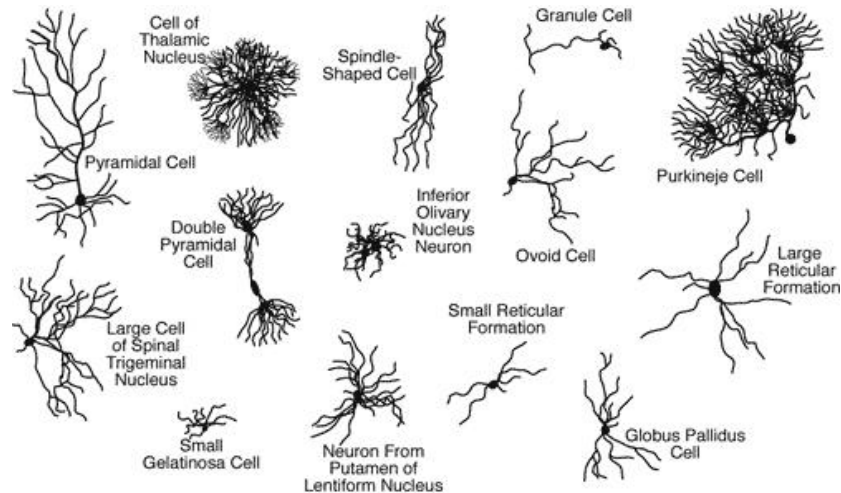
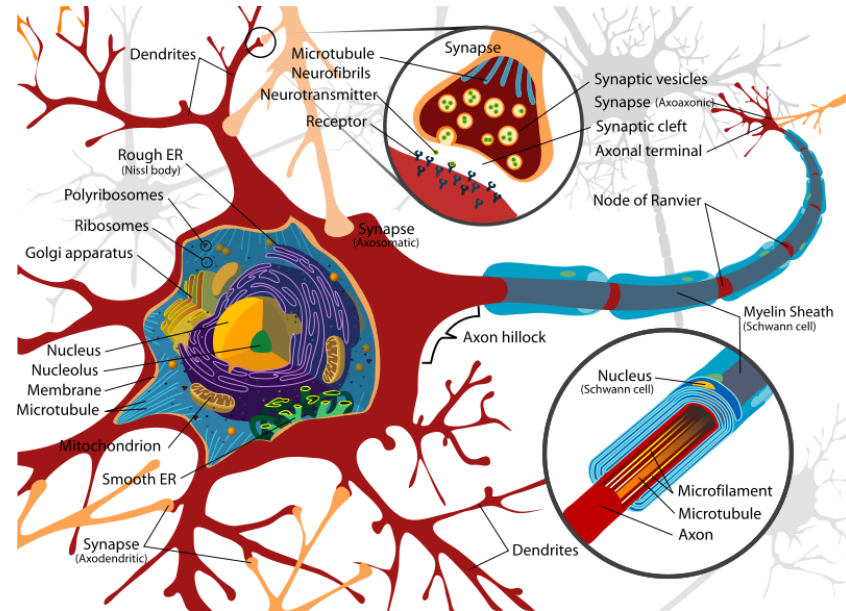
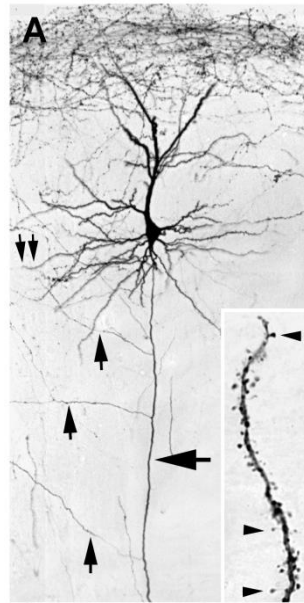
- Examples of tasks that we get computers to learn:

- Predict a category (label) for an image
- Translate text from one language to another
- Predict the trajectory of a moving object

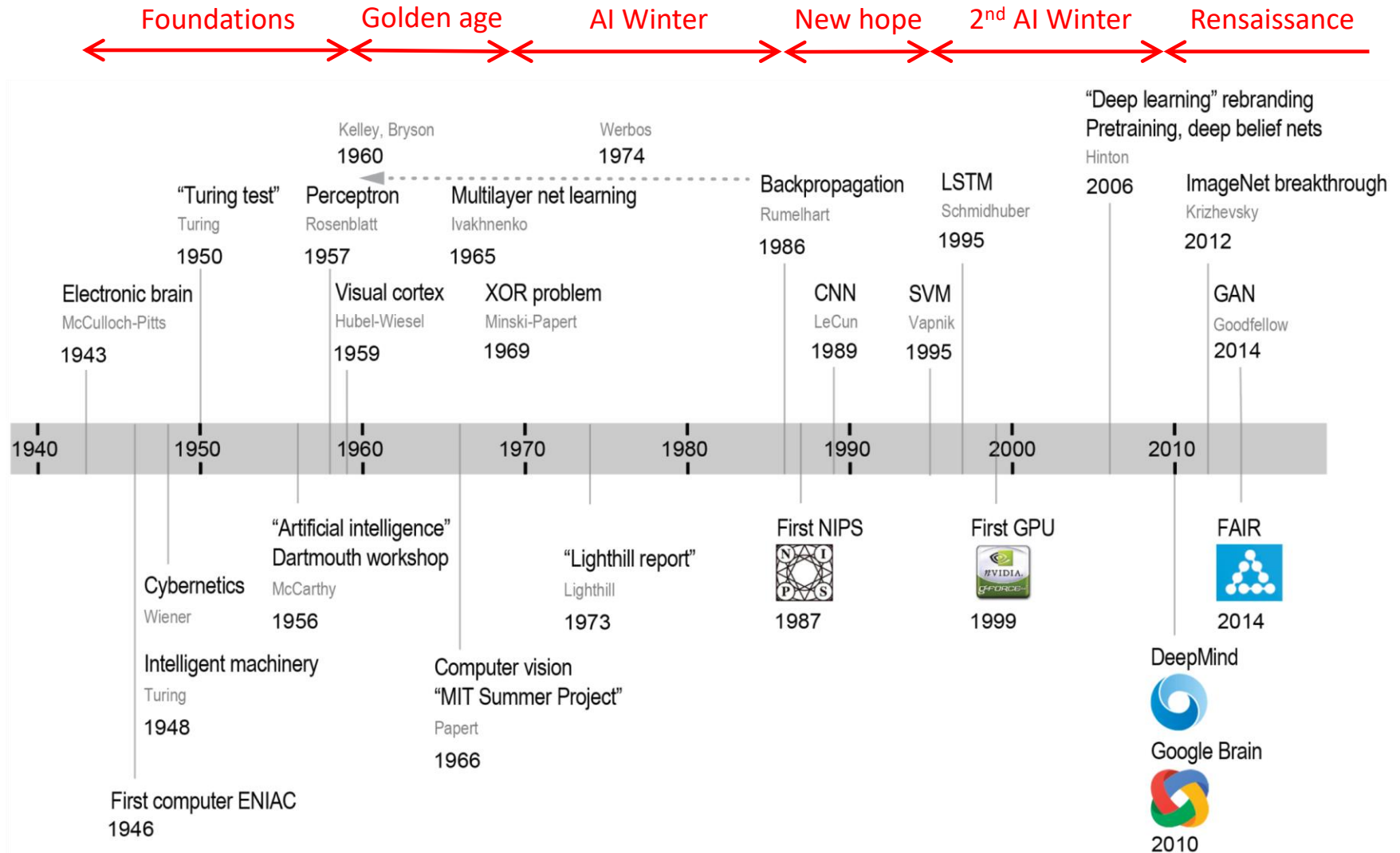
Widely used in the industry

Profound impact in academia (computer vision, NLP)

# Neurons and Neural Networks in Biology



# History



# Content: Today

- ❑ Intro
- ❑ **Machine Learning basics**
  - ❑ Linear regression
  - ❑ Classification
- ❑ Extra: Linear Algebra and Calculus

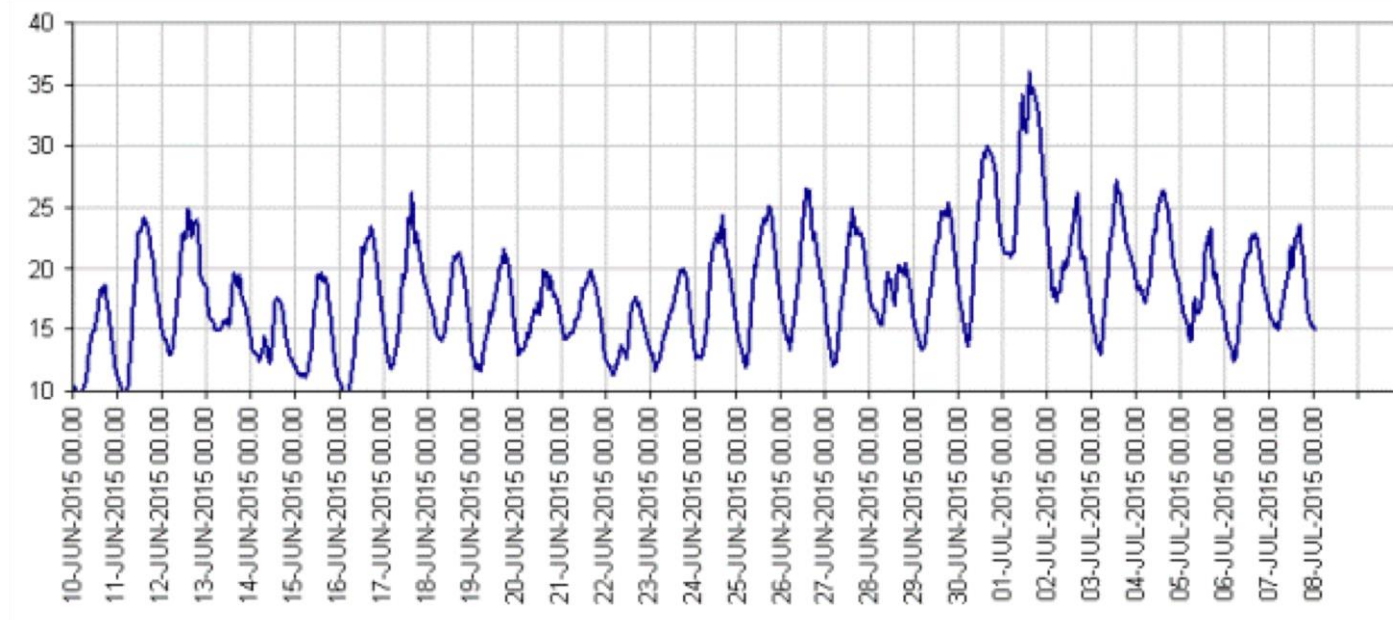
# What will we learn today?

- Supervised machine learning **tasks** of **regression** and **classification**
- How to build **linear models**, i.e. linear parameterisation, to perform those tasks
- How to form an objective function (**loss**) that measures how accurate are our model predictions
- To train machine learning model means to find parameters that explain our training data and can make predictions on new unseen data

# Linear regression



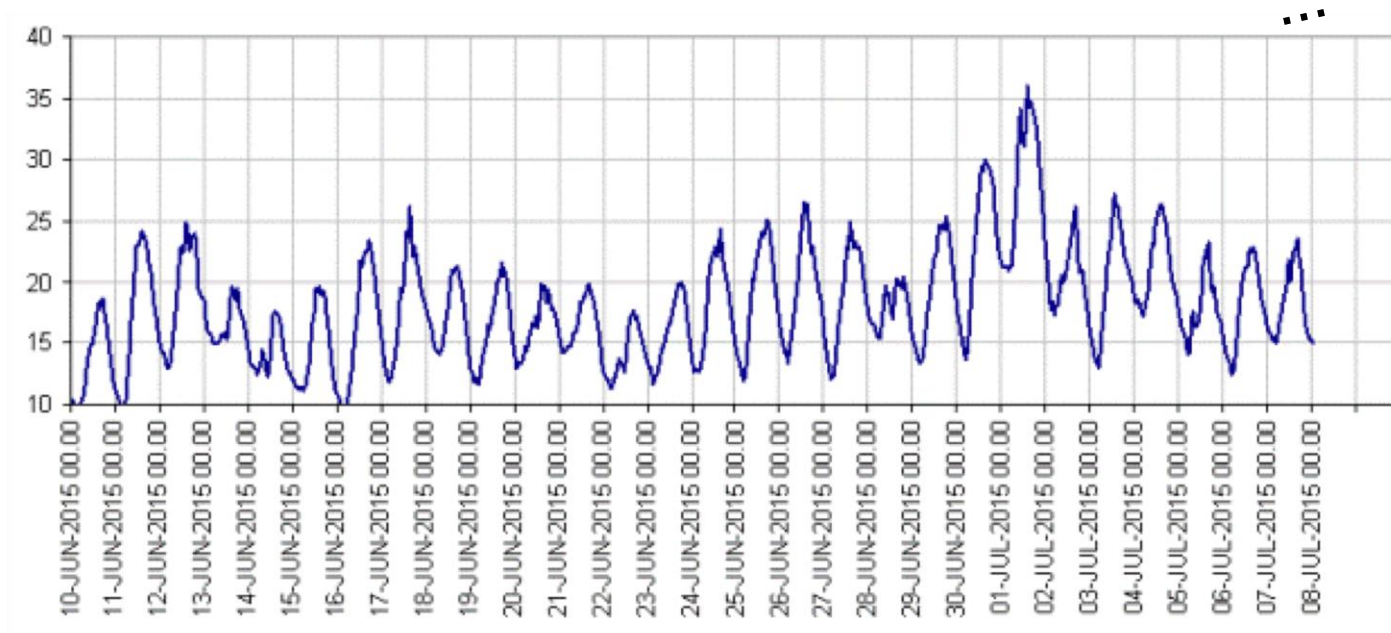
## Example: weather prediction



Forecast air temperature in Brighton

## Example: weather prediction

Rain  
Wind  
Daylight time  
Air pollution  
...



Task: predict air temperature (real number  $\mathbb{R}$  )

Experience: historic data of air temperature (train data)

Performance measure: deviation of the forecast (test data)

# A more formal description of machine learning...

- We have some training data:  $(X, Y)$   
 $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}, \mathbf{x}_i \in \mathbb{R}^D$   
 $Y = \{y_1, y_2, \dots, y_N\}, y_i \in \mathbb{R}$
- Want to obtain a model:  $\hat{y} = f(x, \boldsymbol{\theta}), \boldsymbol{\theta} \in \mathbb{R}^D$
- that approximates the true underlying relationship between X and Y:  $y = F(\mathbf{x})$
- such that  $f(\mathbf{x}_{\text{test}}, \boldsymbol{\theta}) \approx F(\mathbf{x}_{\text{test}})$ , i.e.  $\hat{y}_{\text{test}} \approx y_{\text{test}}$

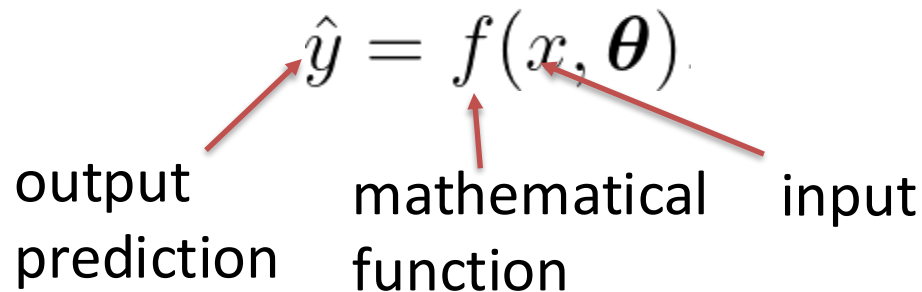
# A more formal description of machine learning...

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**Machine learning = function estimation**

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What type of learning paradigm is this?



Unsupervised learning (A)

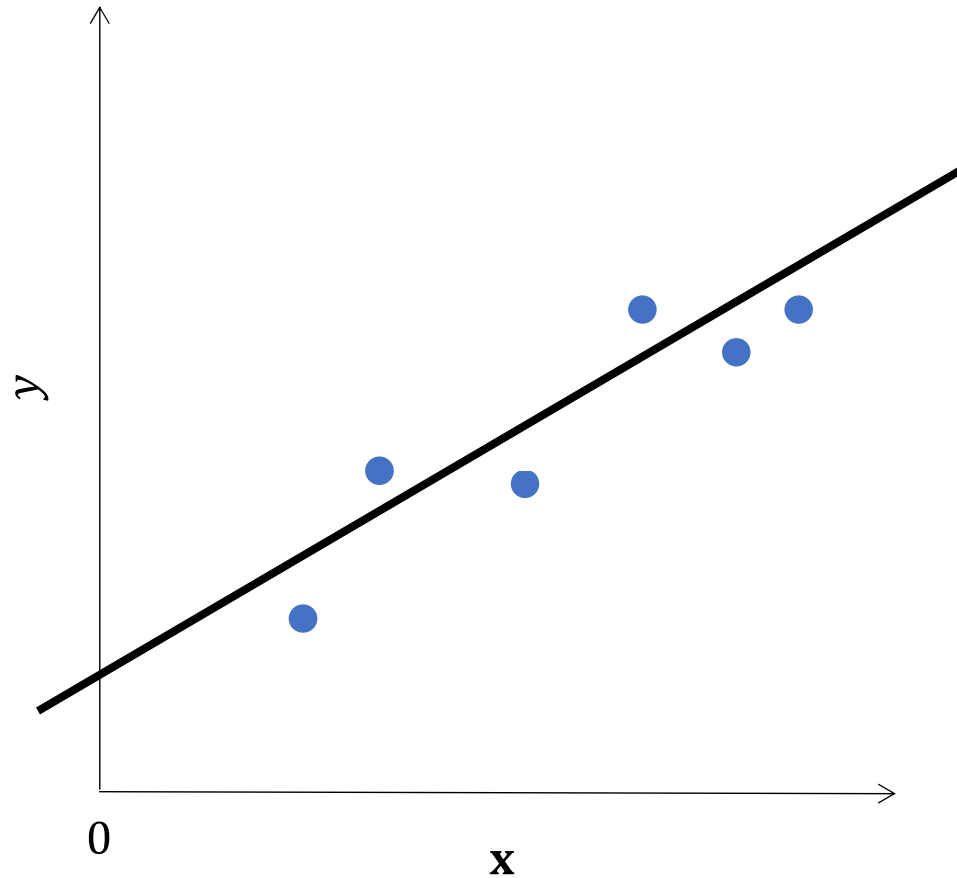
Semi-supervised learning (B)

Supervised learning (C)

Reinforcement learning (D)



# Linear model



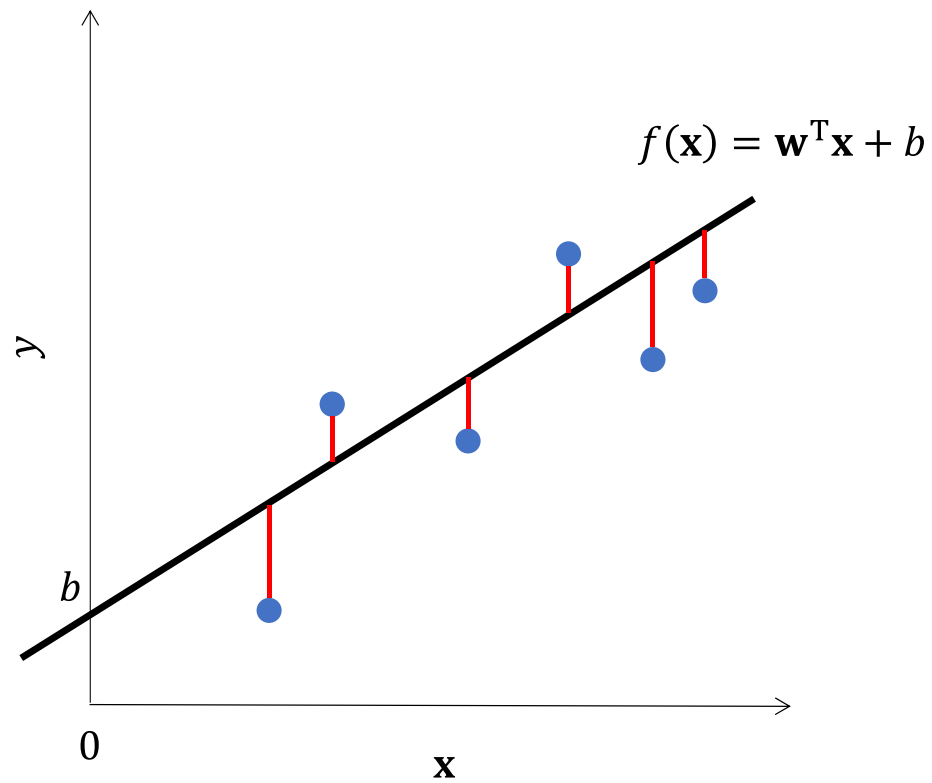
Find linear function to make predictions  $y$

# Linear model

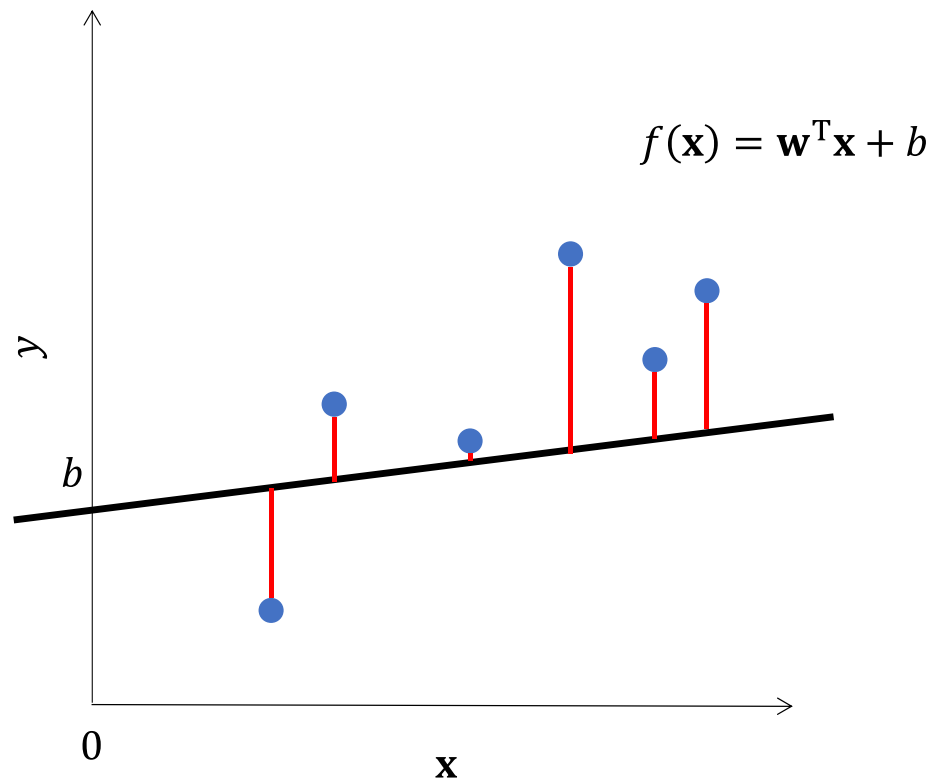
- Given training data  $\{(\mathbf{x}_i, y_i) \sim p \text{ i. i. d.}\}_{i=1}^n$
- Assuming linear model  $\hat{y} = f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$
- Find optimal parameters  $\mathbf{w}, b$  by minimizing empirical loss

$$\hat{L}(\mathbf{w}, b) = \frac{1}{n} \sum_{i=1}^n (\mathbf{w}^T \mathbf{x}_i + b - y_i)^2$$

# Linear model



# Linear model



# Finding optimal parameters

- Matrix-vector notation  $\mathbf{X}\mathbf{w} + b\mathbf{1} = \mathbf{y}$

using the fact that:  $\mathbf{w}^T \mathbf{x} = \mathbf{x}^T \mathbf{w}$

$$\begin{matrix} & d \\ & \mathbf{x}_1^T \\ & \mathbf{x}_2^T \\ n & \mathbf{x}_3^T \\ & \vdots \\ & \mathbf{x}_n^T \end{matrix} \quad \begin{matrix} 1 \\ \mathbf{w} \\ d \end{matrix} + \begin{matrix} 1 \\ b \\ b \\ b \\ \vdots \\ b \\ n \end{matrix} = \begin{matrix} 1 \\ \mathbf{y} \\ n \end{matrix}$$

# Finding optimal parameters

- Matrix-vector notation: absorb the bias into the weight vector

$$\mathbf{X}\mathbf{w} = \mathbf{y}$$

The diagram illustrates the matrix-vector equation  $\mathbf{X}\mathbf{w} = \mathbf{y}$  with dimensions and structure:

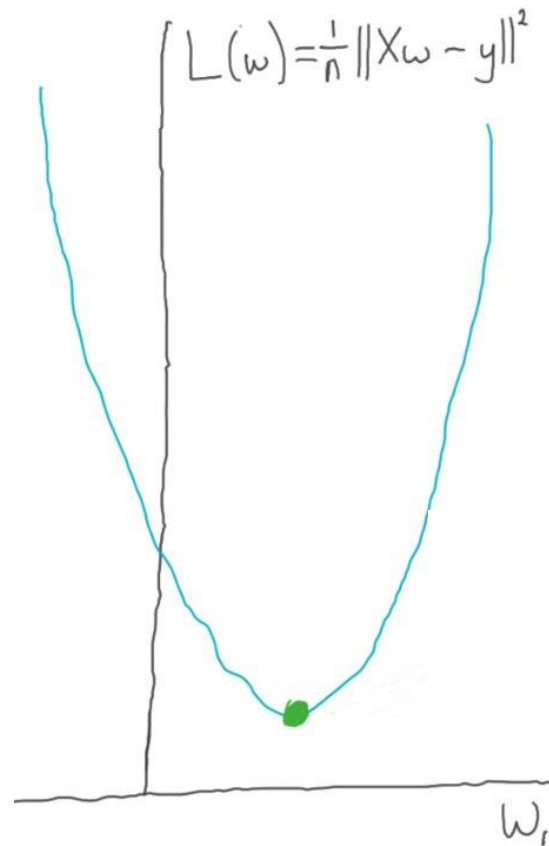
- Matrix  $\mathbf{X}$ :** A matrix with  $n$  rows and  $d + 1$  columns. The rows are labeled  $\mathbf{x}_1^T, \mathbf{x}_2^T, \mathbf{x}_3^T, \vdots, \mathbf{x}_n^T$ . The last column contains the value 1 for each row, representing the bias term.
- Vector  $\mathbf{w}$ :** A vector with  $d + 1$  elements. The top element is labeled  $\mathbf{w}$  and the bottom element is labeled  $b$ , representing the bias.
- Vector  $\mathbf{y}$ :** A vector with  $n$  elements, labeled  $\mathbf{y}$ .

The equation is shown as:

$$\begin{matrix} d + 1 \\ \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \mathbf{x}_3^T \\ \vdots \\ \mathbf{x}_n^T \end{matrix} \begin{matrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{matrix} \begin{matrix} 1 \\ \mathbf{w} \\ b \end{matrix} = \begin{matrix} 1 \\ \mathbf{y} \end{matrix}$$

# Finding optimal parameters

- Loss  $\hat{L}(\mathbf{w}) = \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$





# Finding optimal parameters

- Loss  $\hat{L}(\mathbf{w}) = \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

- Set the gradient to zero to get the minimizer:

$$0 = \nabla_{\mathbf{w}} \hat{L}(\mathbf{w}) = \frac{1}{n} \nabla_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$$

$$0 = \nabla_{\mathbf{w}} [(\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y})]$$

$$0 = \nabla_{\mathbf{w}} [\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y}]$$

$$0 = 2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y}$$

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\|\mathbf{a}\|^2 = \sum_i a_i^2$$

$$\|\mathbf{a}\|^2 = \mathbf{a}^T \mathbf{a} = \langle \mathbf{a}, \mathbf{a} \rangle$$

# Regularization

- Inverse well-defined only when  $\text{rank}(\mathbf{X}^T \mathbf{X}) = d$  implying  $n \geq d$   
= (over-)determined system = more equations than variables
- Otherwise, add **regularization term (weight decay)**

$$\hat{L}(\mathbf{w}) = \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \alpha \|\mathbf{w}\|^2$$

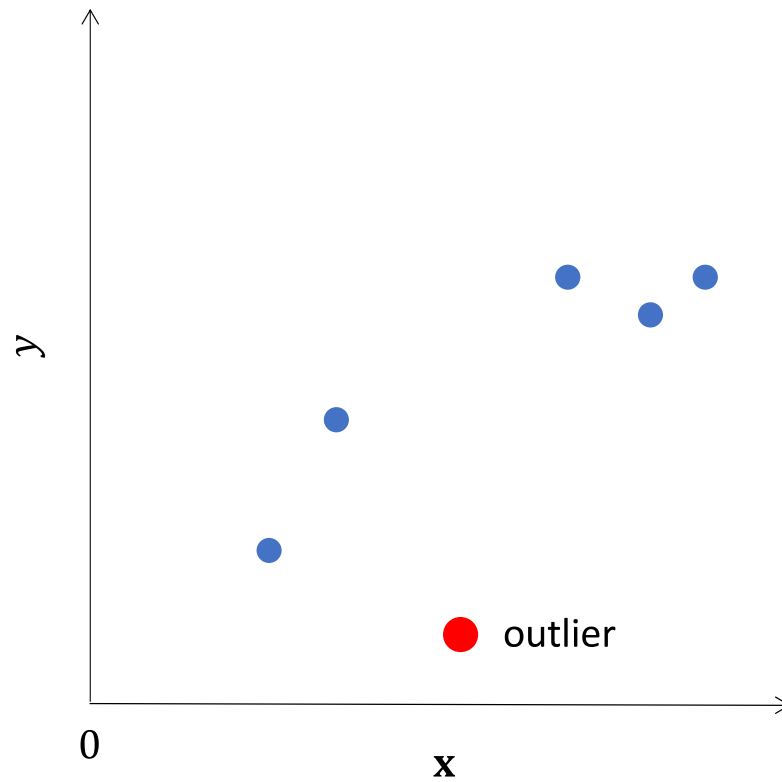
$$0 = \nabla_{\mathbf{w}} \hat{L}(\mathbf{w}) = \frac{2}{n} \mathbf{X}^T \mathbf{X} \mathbf{w} - \frac{2}{n} \mathbf{X}^T \mathbf{y} + 2\alpha \mathbf{w}$$

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X} + n\alpha \mathbb{I})^{-1} \mathbf{X}^T \mathbf{y}$$

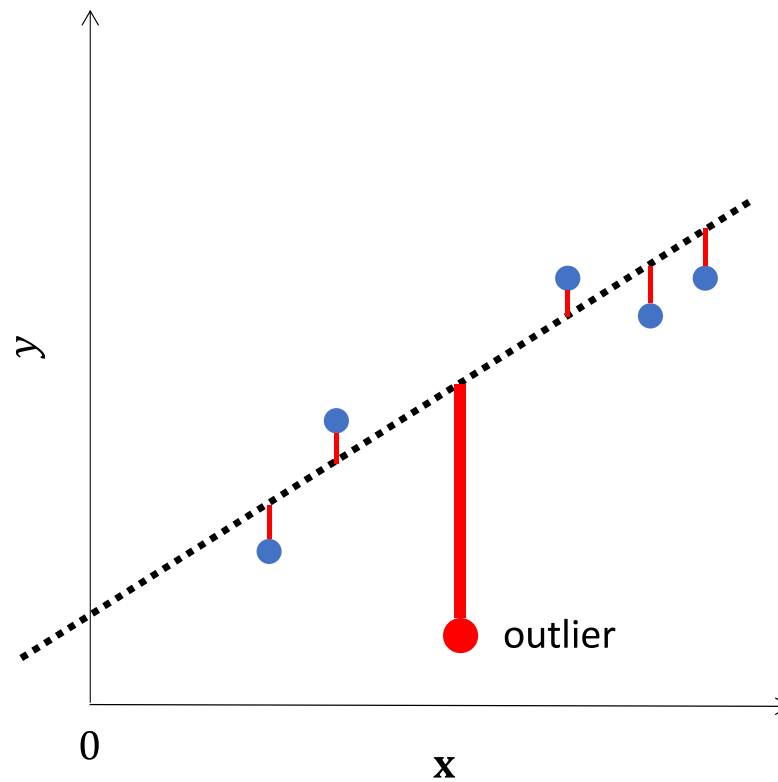
# Loss choice

- $L_2$  loss:  $\ell(\hat{y}, y) = |\hat{y} - y|^2$
- $L_1$  loss:  $\ell(\hat{y}, y) = |\hat{y} - y|$
- 0-1 loss:  $\ell(\hat{y}, y) = \begin{cases} 1 & \hat{y} = y \\ 0 & \hat{y} \neq y \end{cases}$
- many other...

# Loss choice

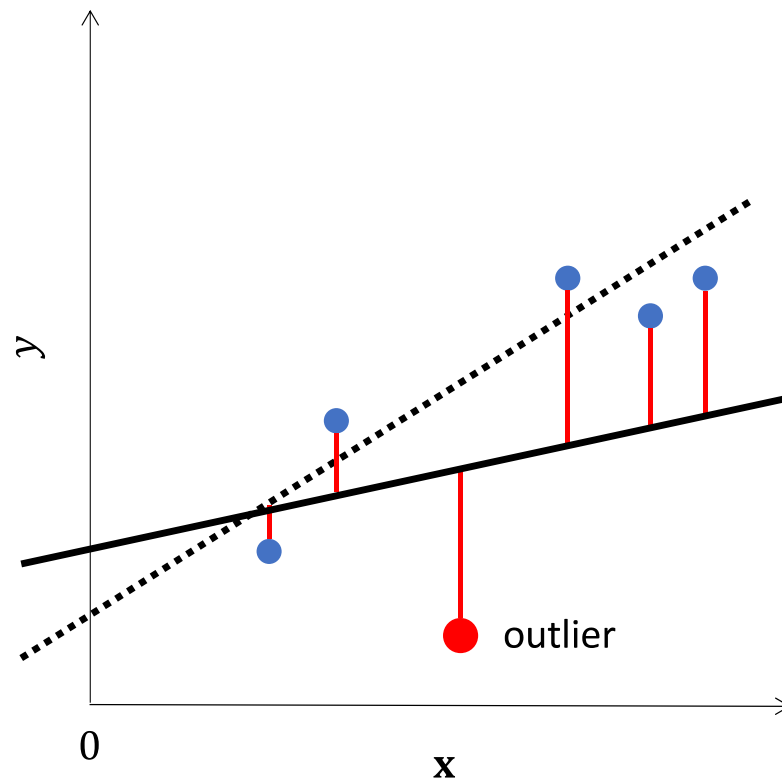


# Loss choice



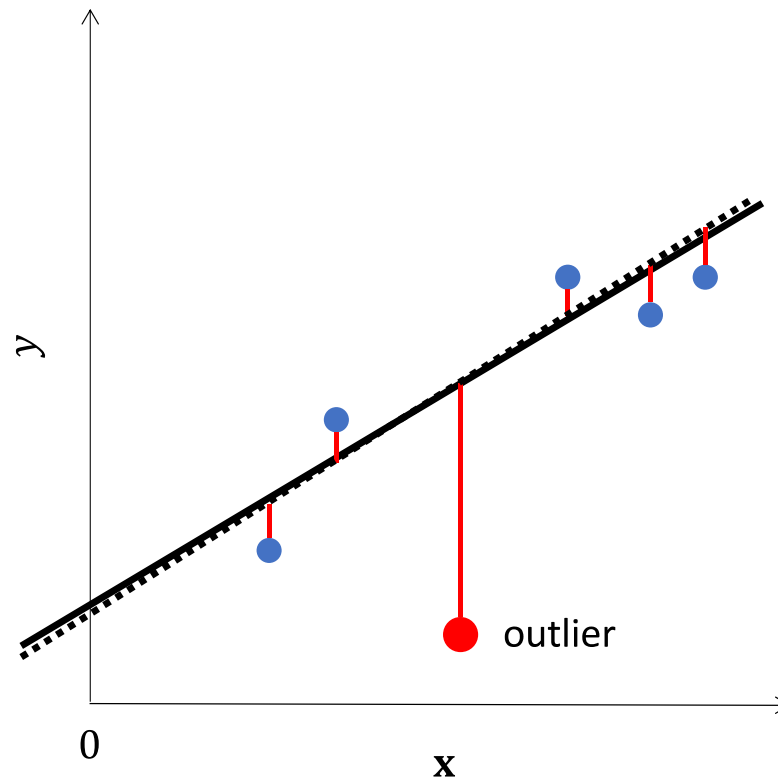
$$L_2 \text{ loss} \quad \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 = \frac{1}{n} \sum_{i=1}^n (\mathbf{w}^T \mathbf{x}_i + b - y_i)^2$$

# Loss choice



$$L_2 \text{ loss } \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 = \frac{1}{n} \sum_{i=1}^n (\mathbf{w}^T \mathbf{x}_i + b - y_i)^2$$

# Loss choice



$$L_1 \text{ loss } \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_1 = \frac{1}{n} \sum_{i=1}^n |\mathbf{w}^T \mathbf{x}_i + b - y_i|$$

# Linearity is in terms of parameters

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b = w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b$$

Models must be linear in the weights,  $\mathbf{w}$ . Non-linear features ( $\mathbf{x}_i$ ) are Allowed.

$$f(x) = w_1 x_1 + b$$

$$f(x) = w_1 x_1 + w_2 x_2^2 + b$$

$$f(x) = w_1 x_1 + w_2 \cos(x_2) + \dots + w_M f_M(x_M)$$

Here indexing is in the dimension of the feature space, i.e.  $\mathbf{x} \in \mathbb{R}^d$



# Short summary

1. We have learned **the task of regression** (predicting real valued output).
2. We have learned **linear model**, where linearity is in terms of parameters  **$\mathbf{w}$** .
3. We have introduced a **loss function** that measures how well the predictions of our model match the (labeled) data. L2 loss in regression measures deviation from the desired output.
4. Training machine learning model means finding optimal parameters ( **$\mathbf{w}$** ) by minimising the loss of mispredictions.
5. In linear regression, we have derived a closed form expression to compute the optimal parameters.

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